

The Modal Triplet Theory Program D2: Matter–Antimatter Orientation and the Baryogenesis Source Contract A Conditional Interpretation of the Finite MTT Carrier

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Version 2

Abstract

This paper re-evaluates the proposal that the cosmic matter–antimatter asymmetry is a discrete orientation selection in Modal Triplet Theory (MTT). The current finite MTT construction supplies an explicit particle–antiparticle carrier and exact representation-theoretic consistency checks. That is substantial structural progress, but it is not a source theorem for a nonzero baryon abundance.

We prove a finite carrier non-selection theorem: the same conjugation-symmetric particle–antiparticle space and the same baryon-sign observable admit a charge-conjugation-symmetric state with zero asymmetry and branch states with either sign of asymmetry. Carrier geometry, anomaly cancellation, and an available CP-odd phase therefore do not select the state or its population weights. We distinguish two lawful continuations. A branch-state route may postulate or derive a selected asymmetric upper state and then prove its descent. A dynamical route must derive baryon- or lepton-number-changing interactions, CP-odd source terms, nonequilibrium transport or an explicitly declared alternative, and washout control from the selected action.

Discrete orientation remains a useful organizing hypothesis: it can label the conjugate sectors and may participate in a future source mechanism. However, the present result is a carrier theorem plus a completion and falsification contract. It does not derive the observed baryon-to-entropy ratio, eliminate the need for baryogenesis dynamics, predict the absence of antimatter domains, or turn anomaly cancellation into a population-selection law.

Revision note for this edition

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Reason. The first edition treated orientation-related survivors as if their unequal global abundance followed from refinement stability, anomaly cancellation, and irreversibility. The current corpus establishes a finite particle–antiparticle carrier, but it does not establish that source map.

Resolution. Version 2 separates carrier, state, dynamics, and observable. It proves the carrier non-selection result, reclassifies orientation selection as a conditional hypothesis, restores the standard baryogenesis obligations, and states exact promotion and falsification gates.

Retained content.

Matter and antimatter may be represented by conjugate orientation sectors of one finite carrier, and a selected orientation variable may be useful in a future physical mechanism.

Open boundary.

No current MTT theorem selects a nonzero baryon-number expectation, an asymmetric branch probability, a baryogenesis source and washout history, or the observed numerical baryon-to-entropy ratio.

How to read this paper

The paper is organized around one distinction:

having matter and antimatter slots is not the same as selecting how much of each slot the universe occupies.

Section 1 places the question at the correct claim tier. Section 2 explains the exact finite carrier and the meaning of orientation. Section 3 proves that the carrier does not select an asymmetric state. Sections 4–6 separate anomaly constraints, CP data, and irreversibility from an actual abundance source. Sections 7 and 8 formulate two possible completion routes and their required calculations. The final sections compare the proposal with baryogenesis, state observational gates, and record what would falsify the MTT interpretation.

The central picture in plain language. A theatre can contain two labelled sections without deciding where the audience sits. Likewise, a finite Hilbert space can contain particle and antiparticle sectors without choosing a density matrix, a cosmological state, or a history measure. Orientation labels the sections. A source theorem must still determine the occupancy and its evolution.

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1 Question, Scope, and Current Tier

The observed universe contains a small but nonzero excess of baryons over antibaryons. In conventional cosmology this is summarized by a normalized quantity such as

$$\eta_B = \frac{n_B - n_{\bar{B}}}{s},$$

where s is the entropy density. Explaining the sign and magnitude of η_B requires more than naming particle and antiparticle states. One must specify the state, the number observable, the dynamics that changes or preserves that number, and the cosmological history in which the expectation is evaluated.

Program D2 asks whether the MTT upper geometry can source those data. The answer is currently tiered:

D2.1. An explicit finite particle–antiparticle carrier exists.

D2.2. Conjugation and orientation data can be represented on that carrier.

D2.3. No selected state or population law producing $\eta_B \neq 0$ has been derived from that structure.

D2.4. No complete same-source baryogenesis or asymmetric-branch calculation has been executed.

This paper therefore does not claim that MTT has solved baryogenesis. It identifies the exact structural foothold, proves the immediate non-selection boundary, and turns the remaining idea into a testable research contract.

2 The Finite Carrier and the Meaning of Orientation

2.1 What the current MTT construction supplies

The current finite-geometry authority supplies a 96-dimensional three-family particle–antiparticle bimodule for

$$\mathcal{A}_F = \mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C}).$$

Its matrix execution verifies the declared algebra action, star structure, real and grading signs, order-zero relation, and a structural order-one condition on the selected channel incidence. This gives a concrete home for particle and antiparticle slots. The unit incidence coefficients are structural witnesses, not measured Yukawa values or cosmic population weights.

2.2 Orientation as a typed label

Definition 1 (Conjugate orientation sectors). Let

$$\mathcal{H}_F = \mathcal{H}_+ \oplus \mathcal{H}_-$$

and let J_F be an antiunitary real-structure map exchanging the two sectors. The signs $+$ and $-$ are called conjugate orientation labels when they distinguish the two sectors and are reversed by J_F .

This definition is intentionally modest. Depending on the realization, the label may encode particle versus antiparticle, a conjugate representation, a sheet orientation, or another discrete involution. Equality of labels is not equality of particles, and the existence of the involution is not yet a cosmological selection law.

2.3 The baryon-sign observable

To ask an abundance question, introduce a self-adjoint observable

$$B_F = \begin{pmatrix} I_+ & 0 \\ 0 & -I_- \end{pmatrix}, \quad J_F B_F J_F^{-1} = -B_F.$$

The expectation in a normalized state ρ is

$$\langle B_F \rangle_\rho = \text{Tr}(\rho B_F).$$

The carrier and B_F make the question well typed. The density operator ρ is still independent data.

3 The Carrier Non-Selection Theorem

Theorem 1 (A conjugate carrier does not select an asymmetry). *Let $\mathcal{H}_F = \mathcal{H}_+ \oplus \mathcal{H}_-$ have nonzero finite dimensions, let J_F exchange the two sectors, and let B_F have eigenvalue +1 on $\mathcal{H}_+$ and -1 on $\mathcal{H}_-$. Then the same fixed data $(\mathcal{H}_F, J_F, B_F)$ admit normalized states ρ_0, ρ_+, ρ_- with*

$$\text{Tr}(\rho_0 B_F) = 0, \quad \text{Tr}(\rho_+ B_F) = 1, \quad \text{Tr}(\rho_- B_F) = -1.$$

Consequently, carrier geometry and conjugation data alone determine neither the sign nor the magnitude of matter-antimatter asymmetry.

Proof. Choose unit vectors $u_\pm \in \mathcal{H}_\pm$ and define

$$\rho_\pm = |u_\pm\rangle\langle u_\pm|.$$

Their expectations are respectively +1 and -1. The state

$$\rho_0 = \frac{1}{2} (|u_+\rangle\langle u_+| + |J_F u_+\rangle\langle J_F u_+|)$$

is normalized and has equal weight in conjugate sectors, hence zero B_F expectation. All three states use the same carrier, involution, and observable. $\square$

What the theorem means. The result does not say that orientation is irrelevant. It says that orientation belongs to the kinematic inventory, while asymmetry belongs to a state or history. A derivation must therefore explain why one state is selected, how its expectation is transported to four-dimensional particle number, and why later dynamics does not erase it.

3.1 A two-state worked example

Take $\mathcal{H}_F = \mathbb{C}^2$, $B_F = \sigma_z$, and let conjugation exchange the standard basis vectors. The maximally mixed state gives $\langle B_F \rangle = 0$. A pure upper-component state gives +1. A diagonal state

$$\rho_p = \begin{pmatrix} p & 0 \\ 0 & 1-p \end{pmatrix}$$

gives $2p-1$, so every value in $[-1, 1]$ occurs without changing any carrier geometry. This is why an orientation label cannot be used as a hidden continuous fit for the observed abundance: the missing object is the source of p .

4 Why Anomaly Cancellation Is Not Population Selection

Gauge and mixed anomalies are consistency conditions on chiral representations and their quantum currents. They determine whether a gauge theory can be consistently quantized; they do not count how many particles occupy a representation in a cosmological state. The anomaly polynomial is fixed by representation data before a density matrix such as ρ_p is chosen.

This distinction blocks a central shortcut in version 1. If an orientation assignment made the quantum gauge theory anomalous, that assignment would be inadmissible as a theory. But once an anomaly-free representation content is fixed, anomaly cancellation does not choose a positive rather than negative expectation of B_F . In the Standard Model, the relevant anomaly cancellations occur across the chiral multiplets of a generation; they are not a cosmological mechanism that deletes antiparticle states.

Remark 1 (Correct use of anomaly data). Anomaly freedom is a gate on a proposed source theory. It may rule out a candidate interaction or representation assignment. Passing the gate leaves the state-selection and transport problem intact.

5 What a CP Phase Can and Cannot Do

A CP-odd phase can enter interference terms and permit different rates for conjugate processes. That makes it relevant to baryogenesis, but not sufficient by itself. The classic Sakharov analysis identifies baryon-number violation, C and CP violation, and departure from thermal equilibrium as the standard conditions for dynamically generating an asymmetry from a symmetric initial state [1]. Electroweak calculations further show that the observed CKM source is not by itself large enough to account for the cosmic asymmetry in the minimal Standard Model [5].

The MTT corpus contains finite-phase and orientation clues associated with the selected q79 branch. At the current authority tier, those clues do not establish a physical map to all CKM angles, a baryon-number source, or η_B . A shared-circle holonomy becomes relevant only after one proves an intertwiner

$$L_{\text{shared}} \longrightarrow \text{physical CP-odd source on } \mathcal{H}_F$$

that preserves the selected connection and action and is evaluated in the same cosmological state. Merely observing that both structures contain a phase is not such a theorem.

6 Irreversibility, CPT, and Branch Selection

6.1 Persistence is not production

Suppose a branch already has $\langle B_F \rangle \neq 0$ and the effective dynamics approximately conserves that expectation after some epoch. Irreversibility or washout suppression can then explain persistence. It cannot explain the original sign or magnitude unless the same mechanism also emits the asymmetric state or source term. Production and persistence are different proof obligations.

6.2 CPT-compatible asymmetric states

An asymmetric cosmological state need not mean that the local action violates CPT. A symmetry of the dynamics need not be a symmetry of a particular state or boundary condition. Conversely, one cannot claim that a vague failure of “global admissibility” selects matter while local CPT remains

intact. The claim requires a constructed local quantum field theory, its CPT map, the selected state, and a proof of how the state descends.

The safe current statement is:

MTT may eventually select a CPT-compatible asymmetric branch state, but no current theorem establishes that selection or its probability.

6.3 Two branches are not automatically two universes

The existence of conjugate mathematical states says only that both are allowed by the carrier. Interpreting them as two physical universes, as two time orientations, or as one realized and one hidden branch requires an additional ontology and selection law. Program D2 does not assume any of those readings.

7 Completion Route A: A Selected Asymmetric Branch State

The first lawful route is kinematic. One selected upper state Ω_{up} may descend to a finite state ρ_F and then to an effective four-dimensional state. To close this route, one must provide:

- A1.** a normalized upper state or history measure;
- A2.** a selected branch rule that does not use the observed asymmetry as its selector;
- A3.** an intertwiner from the upper orientation observable to B_F ;
- A4.** a state-preserving descent to the four-dimensional baryon current;
- A5.** a proof that subsequent evolution preserves the sign and yields a controlled magnitude;
- A6.** a probability or typicality statement if the theory claims that the observed branch is preferred rather than merely allowed.

This route can consistently say that the asymmetry is an initial or global state property. It cannot call that property a derived prediction until the source and branch rule are supplied.

8 Completion Route B: Selected Baryogenesis Dynamics

The second route derives an asymmetry dynamically from a declared earlier state. At effective level, many mechanisms can be organized schematically by a transport equation

$$\dot{n}_B + 3Hn_B = S_B[\Phi_{\text{CP}}, \Gamma_{\Delta B}, T, \dots] - \Gamma_{\text{wash}} n_B.$$

Here S_B is a CP-odd baryon source, $\Gamma_{\Delta B}$ represents baryon- or lepton-number-changing processes, and Γ_{wash} removes the generated asymmetry. This equation is a diagnostic template, not an MTT derivation.

To promote it, MTT must derive from one selected action:

- B1.** the relevant number current and its violation law;
- B2.** the CP-odd invariant and its coupling to the reactions;
- B3.** the nonequilibrium background, or a precisely stated alternative to the usual Sakharov setting;

- B4.** finite-temperature or nonequilibrium rates with uncertainties;
- B5.** washout and spectator processes;
- B6.** the transition from any lepton asymmetry to baryon asymmetry when that route is used;
- B7.** the final η_B prediction without choosing inputs from the target value.

The current open upper-action, nonperturbative QFT, and no-knob Standard Model value blockers sit upstream of this computation. Closing the finite carrier does not bypass them.

9 Relation to the MTT Program

9.1 Program B2 and B3

Program B2 supplies a conditional language for gauge structure; Program B3 supplies a conditional language for discrete survivor structure. D2 may use their objects once their hypotheses are checked, but it cannot infer a population law from the words “gauge” or “discrete.” Representation, state, and abundance remain different types.

9.2 Program D1

Program D1 distinguishes an availability mask from a covariant source. Program D2 has the same logical shape. The particle–antiparticle carrier is the mask of available sectors; the selected state or baryogenesis action is the source. In both papers, the missing physical map cannot be supplied by renaming the structural data.

9.3 The finite Standard Model carrier

The explicit 96-dimensional carrier is the principal new foothold unavailable to version 1. It makes the orientation hypothesis calculable: a future operator can now be written on named particle and antiparticle slots. Its current achievement is exact representation compatibility. Physical finite Dirac entries, full source-owned flavor data, nonperturbative QFT transport, and cosmological state selection remain separate.

10 Observational and Numerical Contract

A completed Program D2 should report at least:

- O1.** the convention for B , L , and $B - L$;
- O2.** the selected initial state and cosmological background;
- O3.** the complete source and washout equations;
- O4.** the predicted sign and interval for η_B ;
- O5.** the relation to light-element abundances and cosmic microwave background constraints;
- O6.** any predicted antimatter-domain or isocurvature signature;
- O7.** an uncertainty budget and a held-out comparison protocol.

The numerical value of η_B may be used for final comparison, not as the coefficient that chooses an orientation, phase, branch, or rate. If one to three physical primitives are retained, they must be declared before the comparison and shown not to be disguised replay coordinates.

11 Falsification and Non-Promotion Rules

The orientation program fails at its claimed tier if any of the following occurs:

- F1.** the proposed orientation operator does not intertwine with the physical particle–antiparticle carrier;
- F2.** the selected upper state descends to zero or the wrong-sign asymmetry;
- F3.** the source action violates gauge consistency, locality, unitarity, or the declared CPT contract;
- F4.** washout removes the generated signal;
- F5.** the calculated asymmetry depends on choosing a branch or coefficient from the observed η_B ;
- F6.** a claimed prediction lacks an uncertainty-controlled comparison with cosmological data.

The following statements are not promotion evidence:

- the carrier contains antiparticle slots;
- a finite phase is numerically near a measured CP coordinate;
- an anomaly-free representation exists;
- one asymmetric state can be written down;
- irreversible evolution preserves an asymmetry inserted by hand.

12 Conclusion

Program D2 now has a firmer and narrower foundation. MTT possesses a concrete finite particle–antiparticle carrier on which orientation, conjugation, and number observables can be represented. The carrier non-selection theorem shows exactly why that achievement does not yet explain the cosmic abundance: the same finite data support symmetric and oppositely asymmetric states.

Discrete orientation therefore survives as a candidate source coordinate, not as a completed explanation. The next scientific advance must select either an asymmetric upper state with a lawful descent or a baryogenesis dynamics with number violation, CP-odd sourcing, nonequilibrium transport, and washout control. Both routes can be tested on the existing finite carrier. Neither is supplied by anomaly cancellation, a phase label, or irreversibility alone.

This correction is constructive rather than destructive. It replaces an unverifiable narrative with a finite operator problem and a clear exit certificate. A successful future version will not merely say that matter is the selected orientation; it will calculate why the selected state has the observed sign and magnitude and show that the result survives the complete four-dimensional dynamics.

Computational Evidence and Reproducibility

The finite carrier result used here is archived in the curated MTT results repository under authority A48:

<https://github.com/PeterNero/mtt-results-repro/tree/main/release/authority/A48>.

Authority A48 establishes the finite representation carrier and explicitly does not promote structural incidence coefficients to physical Yukawa or population values. The no-knob Standard Model value and precision-equivalence frontier remains open under blocker B.SM.02 in the MTT research ledger.

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